

Cellular Network Analysis

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Key Takeaways

1. Raw data was inspected for quality, consistency, and accuracy.
2. KPI distributions were measured against target requirements.
3. Raw data was processed to identify the users with the poorest service.
4. Signal strength data was validated between primary and secondary measurements.
5. KPI-to-KPI correlation reveals negligible linear dependence.
6. Spatial analysis identifies potential cell sites with optimization opportunities.

Introduction

This goal of this project was to attain hands-on radio access network (RAN) engineering experience¹. This report will cover how I used a 5G cellular drive-test dataset and RF engineering principles for

1. Data characterization
2. Measurement validation
3. KPI analysis
4. Spatial analysis
5. Optimization recommendations

Drive-tests are real-world performance measurements of cell tower networks. They are routinely conducted to compare with predicted values for optimization.

Tools used in this analysis include Python utilizing the libraries NumPy, Pandas, Matplotlib, and Seaborn.

Data Characterization

Table 1 provides a quick description of the drive-test scenario.

Area	21 km radius around Patna, Bihar, India
Total # of measurements	16,829
Network types	3G, 4G, LTE, 5G
Test Density	~0.8 km/network type
Time period	36 days at unique times

Table 1: High-level description of scenario

Figure 1 is a sample of the CSV file of the drive-test dataset.

Timestamp	Locality	Latitude	Longitude	Signal Strength (dBm)	Signal Quality (%)	Data Throughput (Mbps)	Latency (ms)	Network Type	BB60C Measurement (dBm)	srsRAN Measurement (dBm)	BladeRFxA9 Measurement (dBm)
2023-05-05 12:50:40.000000	Anisabad	25.599109	85.137355	-84.274113	0.0	1.863890	129.122914	3G	0.000000	0.000000	0.000000
2023-05-05 12:53:47.210173	Fraser Road	25.433286	85.070053	-97.653121	0.0	5.132296	54.883606	4G	-95.810791	-105.452359	-99.920892
2023-05-05 12:56:54.420346	Boring Canal Road	25.498809	85.211371	-87.046134	0.0	1.176985	119.598286	LTE	-91.593861	-95.419482	-87.714070
2023-05-05 13:00:01.630519	Danapur	25.735138	85.208400	-94.143159	0.0	68.596932	46.598387	5G	-90.642773	-101.895905	-96.570698
2023-05-05 13:03:08.840692	Phulwari Sharif	25.538556	85.159860	-94.564765	0.0	38.292038	30.342828	5G	-90.489100	-103.318304	-95.102467

Figure 1: First 5 rows of the data se

Table 2 is a dictionary of measurements.

Metric	Role
Timestamp	Time of data acquisition
Locality	Location of data acquisition
Latitude	Coordinates
Longitude	“”
Signal Strength (dBm)	Primary RF power measurement aka SS-RSRP
Signal Quality (%)	signal strength relative to the maximum possible signal strength ¹
Data Throughput (Mbps)	Capacity of network data transmission
Latency (ms)	Time delay between the transmission and reception of data packets ¹
Network type	Standards and requirements for each network’s measurements
BB60C (dBm)	Secondary signal strength measurement from spectrum analyzer
srsRAN (dBm)	Secondary signal strength measurement from software-defined radio device
BladeRFxA9 (dBm)	“”

Table 2: Data dictionary

¹Data was synthetically generated using a random uniform distribution within appropriate ranges.

Data Limitations

The dataset was examined to understand the limited inferences and the recommendations I can make.

Table 3 outlines missing pertinent data for a network optimization assessment.

Missing Data	Role
SS-SINR (dB)	Used to assess signal quality and load conditions. It helps distinguish between a strong signal that is heavily interfered with versus a clean signal
Cell IDs	Antenna or environmental issues may be addressed
Handover success rate (%)	Indicates success rate of users moving between cells without failure
Drop rate (%)	Indicates rate of active sessions being abnormally terminated before the user explicitly terminates them

Table 3: Missing data for consideration

KPI Analysis

Figure 2 illustrates the distribution of 5G KPIs (IQR: 25-75%, median: 50%). The signal strength (SS-RSRP) targets are approximately ≥ -105 dBm. Signal Quality (SS-RSRQ) targets are app. ≥ -12 dBm. Throughput targets are app.

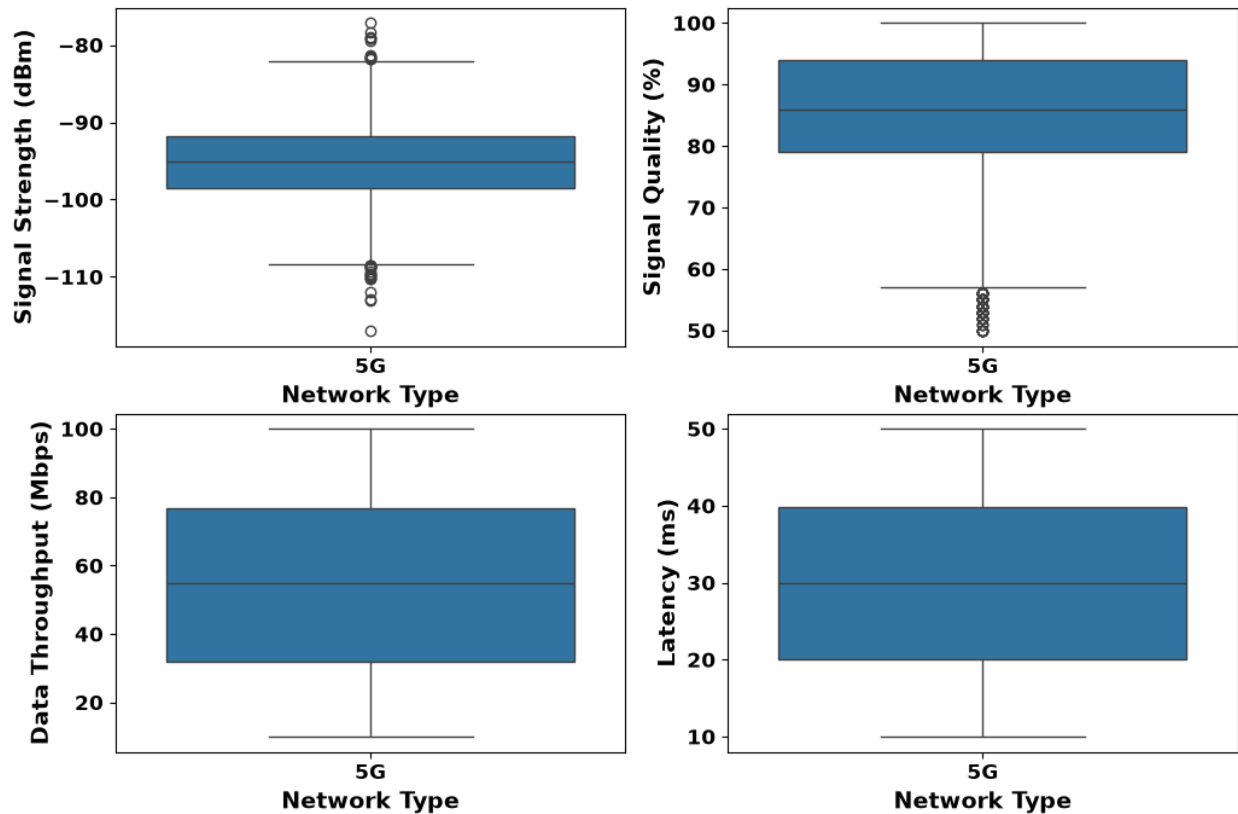


Figure 2: Signal Strength Distribution

Quality of experience (QoE) is a metric intended to position the user's experience at the forefront by blending all the KPIs together. This calculation was performed for every single test. The applied weighting factors were chosen to favor. The process of manipulating the raw KPI data in QoE begins with normalizing the data,

```
KPI.normalized = (KPI - KPI.min_val) / (KPI.max - KPI.min)
```

The latency values were reflected so that a higher value corresponded to a better latency.

Second, the Radio Quality Index (RQI) was calculated from the sum of the weighted normalized radio KPIs, signal quality and signal strength.

```
RQI = 0.6*Signal.Quality.norm + 0.4*Signal.Strength.norm
```

Third, the Service Performance Index (SPI) was calculated from the sum of the weighted normalized performance KPIs, data throughput and latency.

$$SPI = 0.6 * \text{Throughput.norm} + 0.4 * \text{Latency.norm}$$

Finally, the QoE was calculated by the weighted sum of the RQI and SPI and the distribution is shown in Figure 3.

$$QoE = (0.6 * SPI + 0.4 * RQI) * 100\%$$

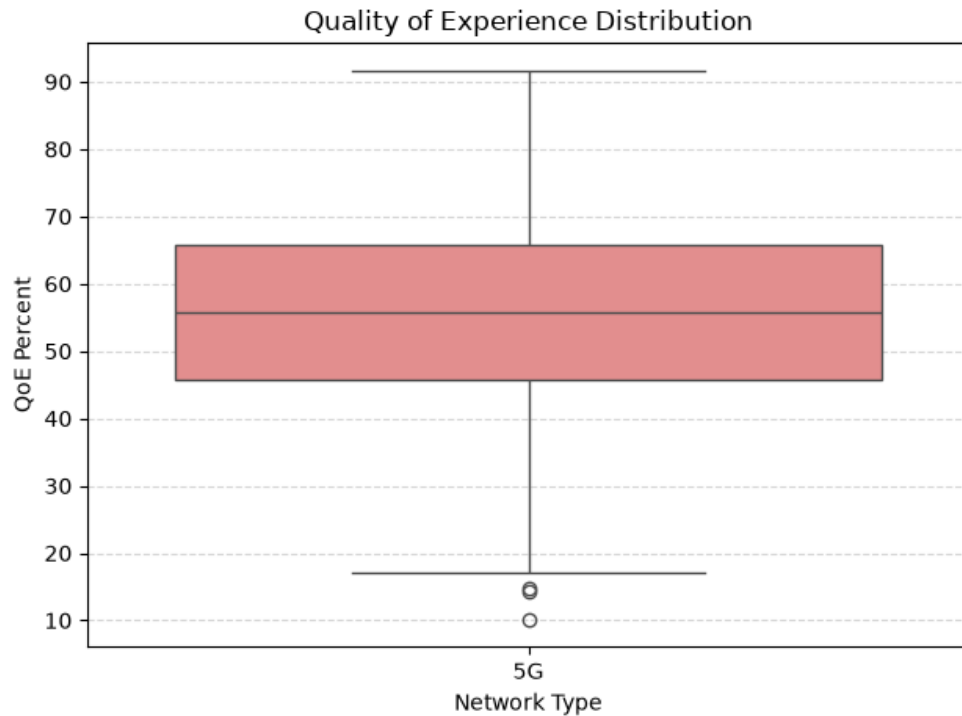


Figure 3: Distribution of QoE for 5G tests

In Table 4 the QoE scores were broken down into 5 ratings to prioritize the most underserved “users.”

QoE Score (%)	QoE Rating
>90	Excellent
80-89	Very Good
70-79	Good
60-69	Fair
0-59	Poor

Table 4: QoE rating categories

Figure 4 scores the correlation of metrics in the 5G population with Poor QoE ratings. The highest scores (closest to +1) occur when the signal strengths are paired with the BB60C,

srsRAN, and BladeRFxA9 measurements. This provides assurance that our signal strength values are very accurate. However, all other correlations are near zero therefore we cannot infer where network issues are occurring in the link.

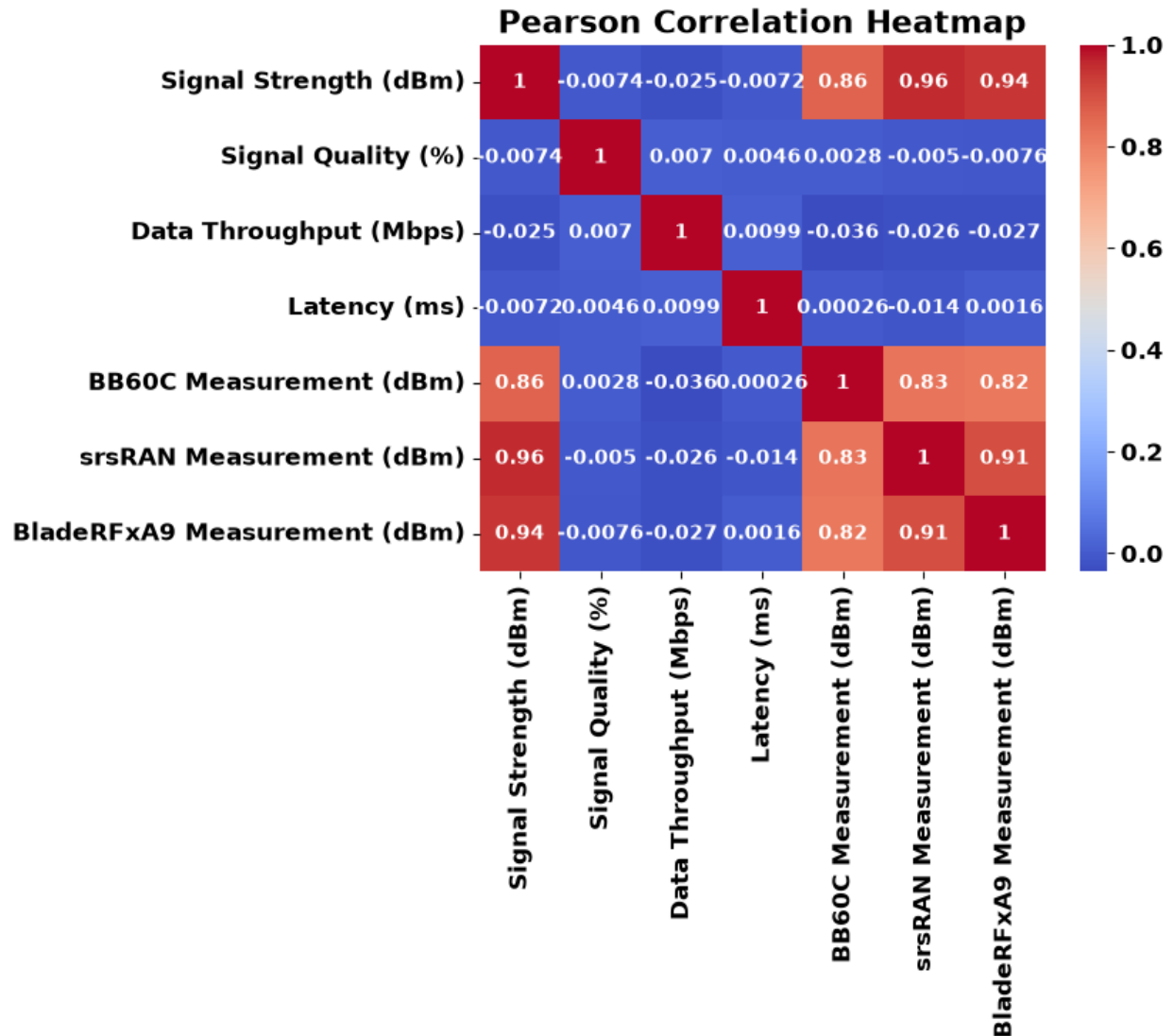


Figure 4: Examining correlations between metrics in the 5G network with Poor QoE ratings

Spatial Analysis

Heatmaps were produced to identify potential trends in network performance. Figure 5 illustrates the spatial distribution of Poor QoE ratings.

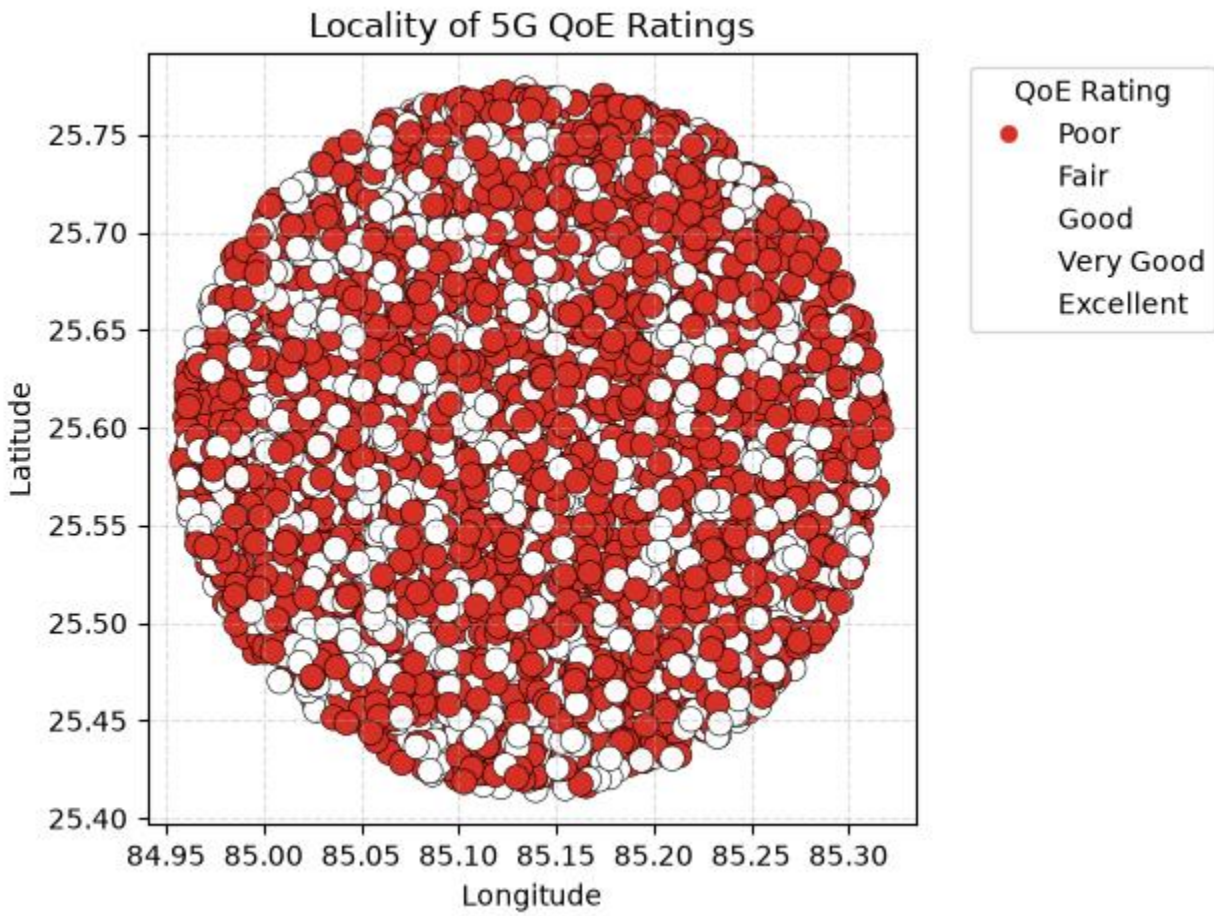


Figure 5: Locality of 5G tests rated a Poor QoE

Optimization Recommendations

Table 5 displays how one may interpret relationships between KPIs.

KPI relationship	Interpretation
Weak signal + low throughput	Coverage limitation
Strong signal + low throughput	Capacity limitation or scheduler/backhaul issue
Strong signal + high latency	Possible congestion or transport delay
Weak signal + high latency	Radio degradation affecting service quality
Excellent signal + excellent throughput	Healthy coverage and capacity

Table 5: Guide to interpret KPI relationships

Table 6 provides a mapping from the problem, to symptom, to root cause, to a corrective action ².

Problem	Symptom (KPIs)	Root Cause	Corrective Action
Coverage Hole	Low RSRP (<-110), high drop rate in area	Terrain blockage, insufficient site density, excessive tilt	Reduce tilt, add repeater/small cell, adjust azimuth toward gap
Overshooting	Serving cell detected far beyond planned radius, high HO failures	Tilt too low, antenna too high, flat terrain	Increase tilt by 2–4°, reduce RS power, add mechanical tilt
Pilot Pollution	High RSRP but low SINR, many cells detected (>5), high HO rate	Multiple strong cells overlap, no dominant server	Tilt strongest interferers, adjust azimuths for clear dominance
Cross-Border Leakage	Signal detected across national/regional boundary	Overshooting toward border, flat terrain	Increase tilt, reduce power, add M-tilt, adjust azimuth
Indoor Coverage Gap	Low RSRP indoor, high indoor drop rate	Building penetration loss exceeds MAPL margin	Deploy indoor DAS/small cells, use low-band SUL carrier

Table 6: Problem to corrective action Matrix

References

- (1) *5G RAN explained: what, how and where next - Ericsson.*
<https://www.ericsson.com/en/ran> (accessed 2026-07-21).
- (2) *4G & 5G RF Planning — A Complete Engineer's Guide...* <https://www.cafetele.com/rf-planning-book/reader.html#ch21> (accessed 2026-07-21).